MATH OPERATIONS

abs	absolute value
acos	arc cosine
asin	arc sine
atan	arc tangent
atan2	arc tangent of quotient
cabsf	complex absolute value
cexpf	complex exponential
cos	cosine
cosh	hyperbolic cosine
cot	cotangent
div	division
exp	exponential
fmod	modulus
log	natural logarithm
log10	base 10 logarithm
matadd	matrix addition
matmul	matrix multiplication
pow	raise to a power
rand	random number generator
sin	sine
sinh	hyperbolic sine
sqrt	square root
srand	random number seed
tan	tangent
tanh	hyperbolic tangent

PROGRAM CONTROL

abort	abnormal program end
calloc	allocate / initialize memory
free	deallocate memory
idle	processor idle instruction
interrupt	define interrupt handling
poll_flag_in	test input flag
set_flag	sets the processor flags
timer_off	disable processor timer
timer_on	enable processor timer
timer_set	initialize processor timer

CHARACTER & STRING MANIPULATION

atoi	convert string to integer
bsearch	binary search of array
isalnum	detect alphanumeric character
isalpha	detect alphabetic character
iscntrl	detect control character
isdigit	detect decimal digit
isgraph	detect printable character
islower	detect lowercase character
isprint	detect printable character
ispunct	detect punctuation character
isspace	detect whitespace character
isupper	detect uppercase character
isxdigit	detect hexadecimal digit
memchr	find first occurrence of char
memcpy	copy characters
strcat	concatenate strings
strcmp	compare strings
strerror	get error message
strlen	string length
strncmp	compare characters
strrchr	find last occurrence of char
strstr	find string within string
strtok	convert string to tokens
system	sent string to operating system
tolower	change uppercase to lowercase
toupper	change lowercase to uppercase

SIGNAL PROCESSING

a_compress	A-law compressing
a_expand	A-law expansion
autocorr	autocorrelation
biquad	biquad filter section
cfftN	complex FFT
crosscorr	cross-correlation
fir	FIR filter
histogram	histogram
iffN	inverse complex FFT
iir	IIR filter
mean	mean of an array
mu_compress	mu law compression
mu_expand	mu law expansion
rfftN	real FFT
rms	rms value of an array

TABLE 29-3
C library functions. This is a partial list of the functions available when C is used to program the Analog Devices SHARC DSPs.

DSPs. A tremendous advantage of using C is the library of functions, both standard C operations, as well as DSP algorithms. Table 29-3 shows a partial list of the C library functions for the SHARC DSPs. The math group includes many functions common in DSP, such as the trig functions (sin, cos, tan, etc.), logarithm, and exponent. If you need these type of functions in your program, this is probably enough motivation in itself to use C instead of assembly. Pay special attention to the "signal processing" routines in Table 29-3. Here you will find key DSP algorithms, including: real and complex FFTs, FIR and IIR filters, and statistical functions such as the mean, rms value, and variance. Of course, all these routines are written in assembly, allowing them to be very efficient in both speed and memory usage.